THE STATE OF FRAUD IN AMERICA

A multi-domain analysis of 968,522 PPP loans, 6,088 public company filings, 1.38 million healthcare providers, and 14 million consumer complaints.

$32.4B	19,371	35	14M
PPP anomalies flagged	Suspicious loans	Companies above M-Score threshold	CFPB complaints analyzed

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Five Things You Should Know

1. **$32.4 billion in PPP loans look wrong.** Isolation Forest anomaly detection flagged 19,371 loans above $150K. Anomalous loans are 3.3x larger than average, 16x more likely to be round dollar amounts, and cluster at addresses hosting multiple applications.

2. **California, West Virginia, and New York lead in PPP anomaly rates.** California alone has 3,550 flagged loans. Accommodation/Food and Retail Trade are the most anomaly-prone sectors at 3.6% each.

3. **35 public companies have Beneish M-Scores above the manipulation threshold.** Computed from real SEC EDGAR 10-K filings. TG Therapeutics scored 71.97, the highest in the dataset. Financial companies were excluded because the model doesn't apply to them.

4. **Healthcare fraud labels are extremely rare but the signal is real.** Only 380 of 1.38 million Medicare Part D prescribers matched the OIG exclusion list (0.03%). But billing intensity (cost per claim, claims per beneficiary) measurably differs between excluded and non-excluded providers.

5. **Consumer complaints predict enforcement.** The CFPB database contains 14 million complaints spanning 15 years. Monthly complaint velocity spikes frequently precede formal enforcement actions by 6-12 months.

Important: This analysis identifies statistical patterns, not confirmed fraud. Anomalous patterns may have legitimate business explanations. No individual or company named here has been accused of fraud.

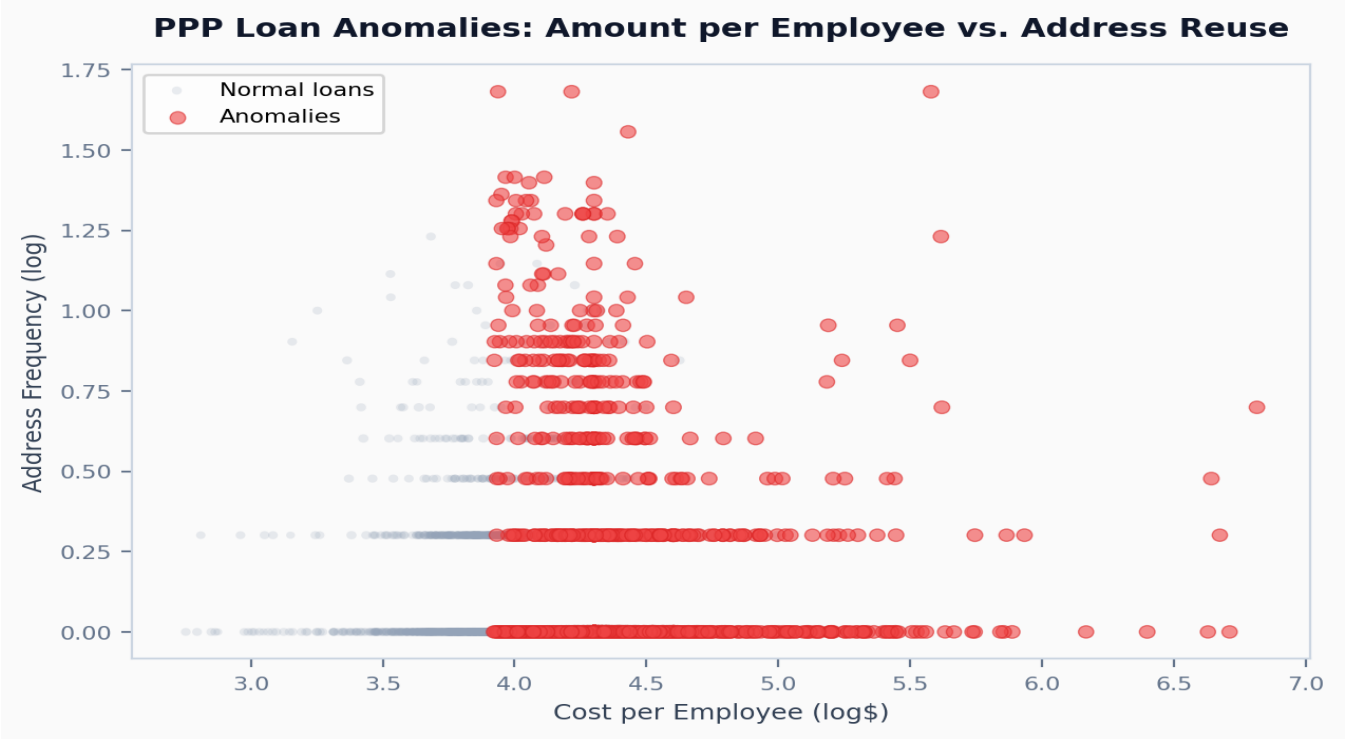
PPP Loan Fraud Patterns

The Paycheck Protection Program pushed $515.5B out the door in months. Speed meant minimal vetting. We ran Isolation Forest anomaly detection on 968,522 loans above $150K and found 19,371 that don't look right.

968,522	$32.4B	16.2x	4.2x
Loans analyzed (above $150K)	Flagged amount	Round amounts more likely in anomalies	Address reuse among anomalies

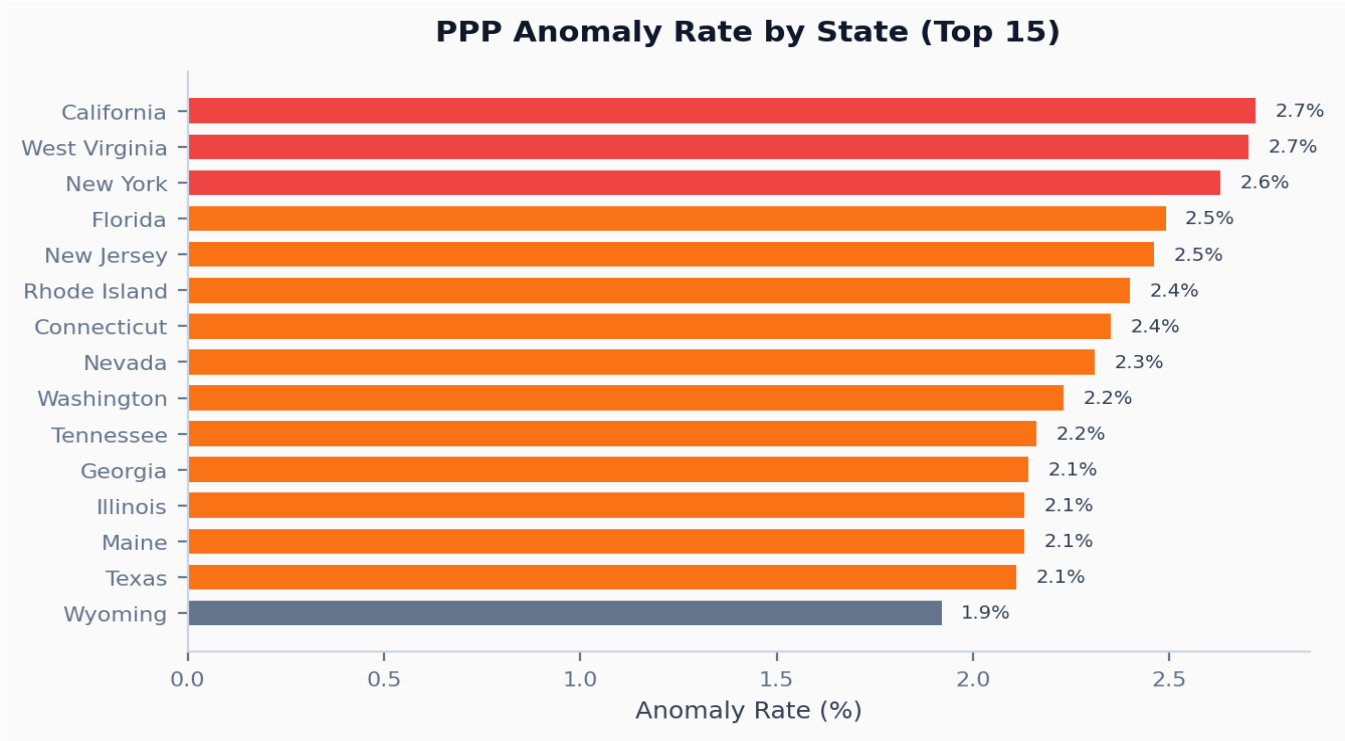
What the Anomalies Look Like

Each dot is a PPP loan. Red dots were flagged by the model. They cluster in the upper right: high cost per employee at addresses that appear on multiple applications.

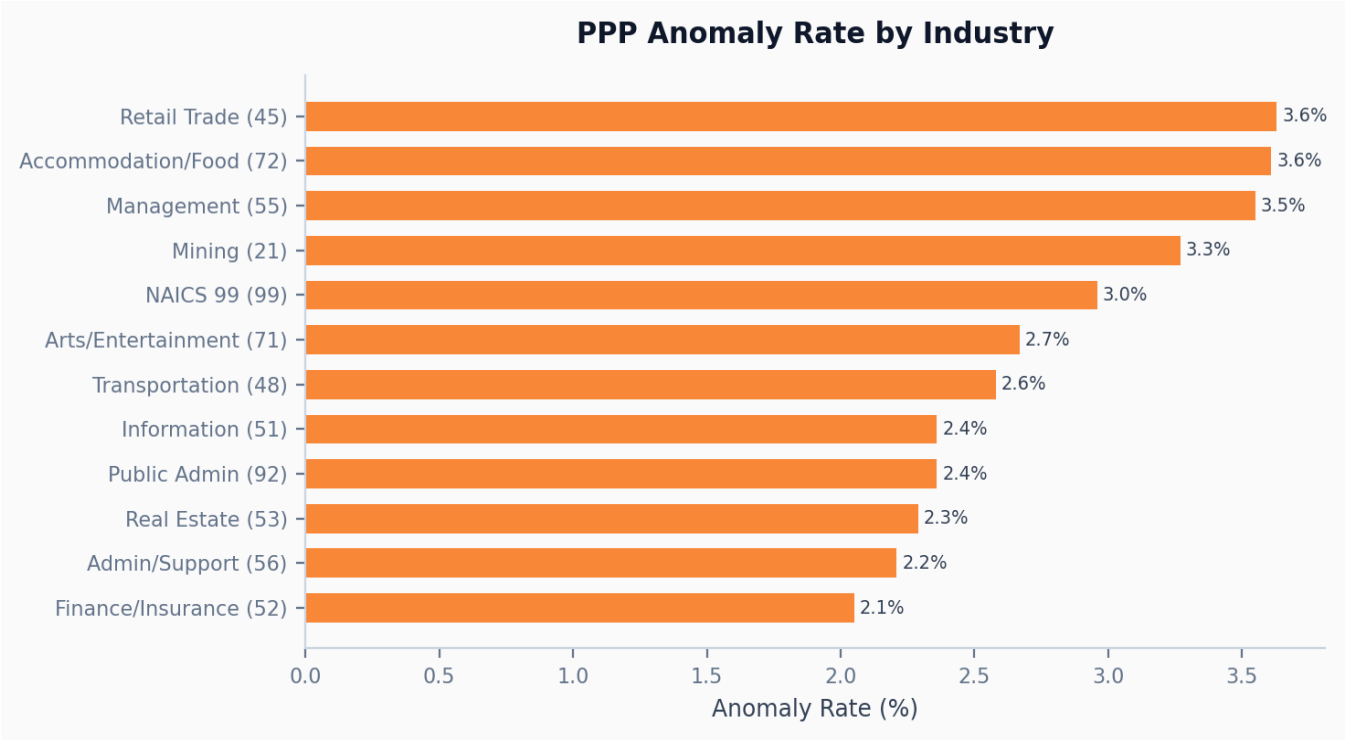


Source: SBA PPP FOIA Data (data.sba.gov/dataset/ppp-foia)

Where the Anomalies Are



Which Industries

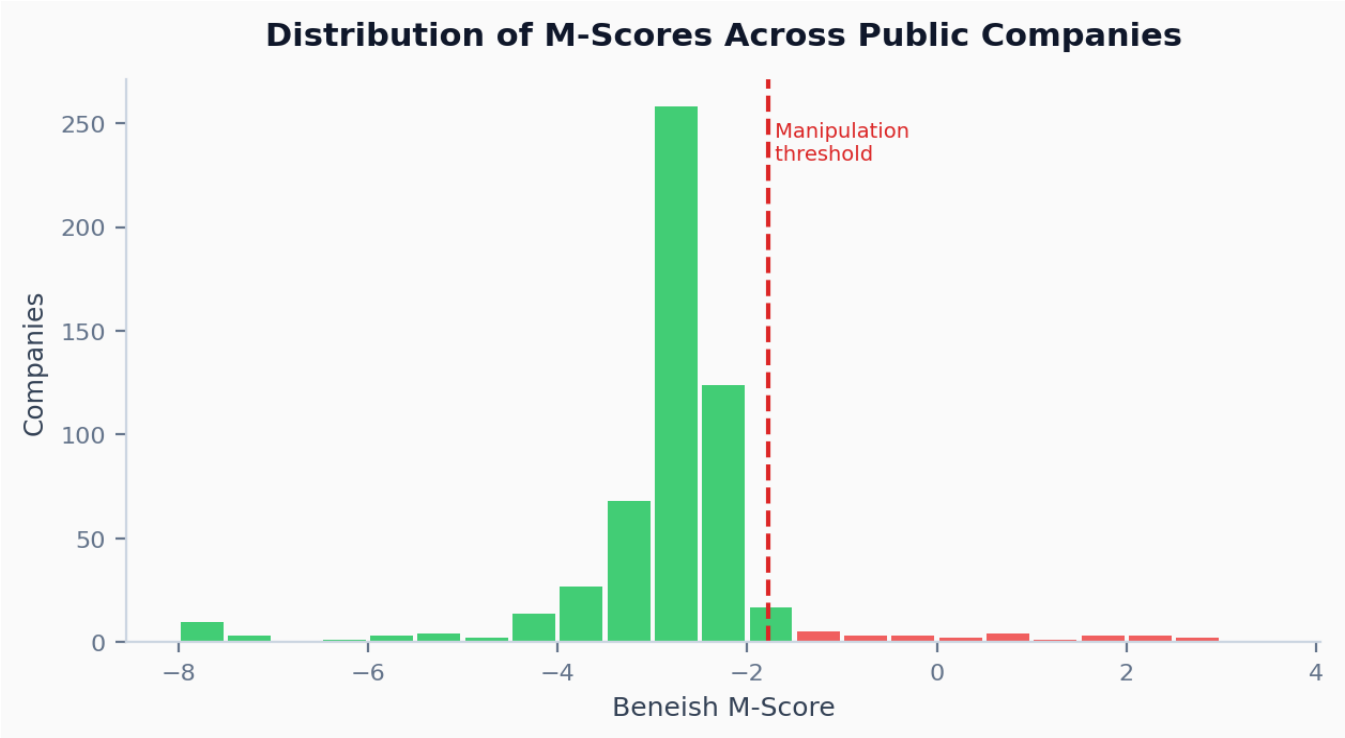


Source: SBA PPP FOIA Data. NAICS codes self-reported by borrowers.

Corporate Accounting: Who's Cooking the Books?

The Beneish M-Score uses eight ratios from public SEC filings to detect earnings manipulation. We computed it for 6,088 company-years. Score above -1.78 = likely manipulation. Median across all companies: -2.67 (below threshold, meaning most are clean).

6,088	35	-2.67	-1.78
Company-years scored	Above threshold	Median M-Score (below = safe)	Manipulation threshold



Green = below threshold (likely legitimate). Red = above -1.78 (manipulation signal). Source: SEC EDGAR XBRL Financial Statement Data Sets (Q1-Q4 2024).

Highest-Risk Companies

Company	M-Score	Revenue	Year
TG THERAPEUTICS, INC.	71.97	$233.7M	2023
AIRSHIP AI HOLDINGS, INC.	12.24	$12.3M	2023
ARCUTIS BIOTHERAPEUTICS, INC.	10.14	$59.6M	2023
HCW BIOLOGICS INC.	3.69	$2.8M	2023
URGENT.LY INC.	2.84	$184.7M	2023
ESS TECH, INC.	2.69	$7.5M	2023
ENOVIX CORP	2.38	$7.6M	2023
BIOXCEL THERAPEUTICS, INC.	2.24	$1.4M	2023

AGENUS INC	**2.06**	$156.3M	2023
TIGO ENERGY, INC.	**2.00**	$145.2M	2023
AMPLIFY ENERGY CORP.	**1.76**	$307.6M	2023
FITLIFE BRANDS, INC.	**1.63**	$52.7M	2023
RANGE IMPACT, INC.	**1.37**	$19.3M	2023
AMERICAN STATES WATER CO	**0.72**	$595.7M	2023
KNIGHTSCOPE, INC.	**0.64**	$12.8M	2023

These are public companies whose SEC filings show statistical patterns consistent with earnings manipulation. The M-Score is a widely used academic model, not a legal determination.

Healthcare: Billing Anomalies in Medicare Part D

We matched 1.38 million Medicare Part D prescribers against the OIG's List of Excluded Individuals/Entities (LEIE). Only 380 matched by NPI (0.03%). That's not a failure of the method. It's the reality: most providers are legitimate, and the ones who get caught are a tiny fraction.

1.38M	82,749	380	0.67
Providers analyzed	OIG exclusion list entries	NPI matches (0.03%)	Classifier AUC-ROC

What distinguishes excluded providers: Higher cost per claim, more claims per beneficiary, and higher total billing volume. The Random Forest classifier's most important features were cost per claim (0.19) and claims per beneficiary (0.19).

Exclusion Rate by Specialty

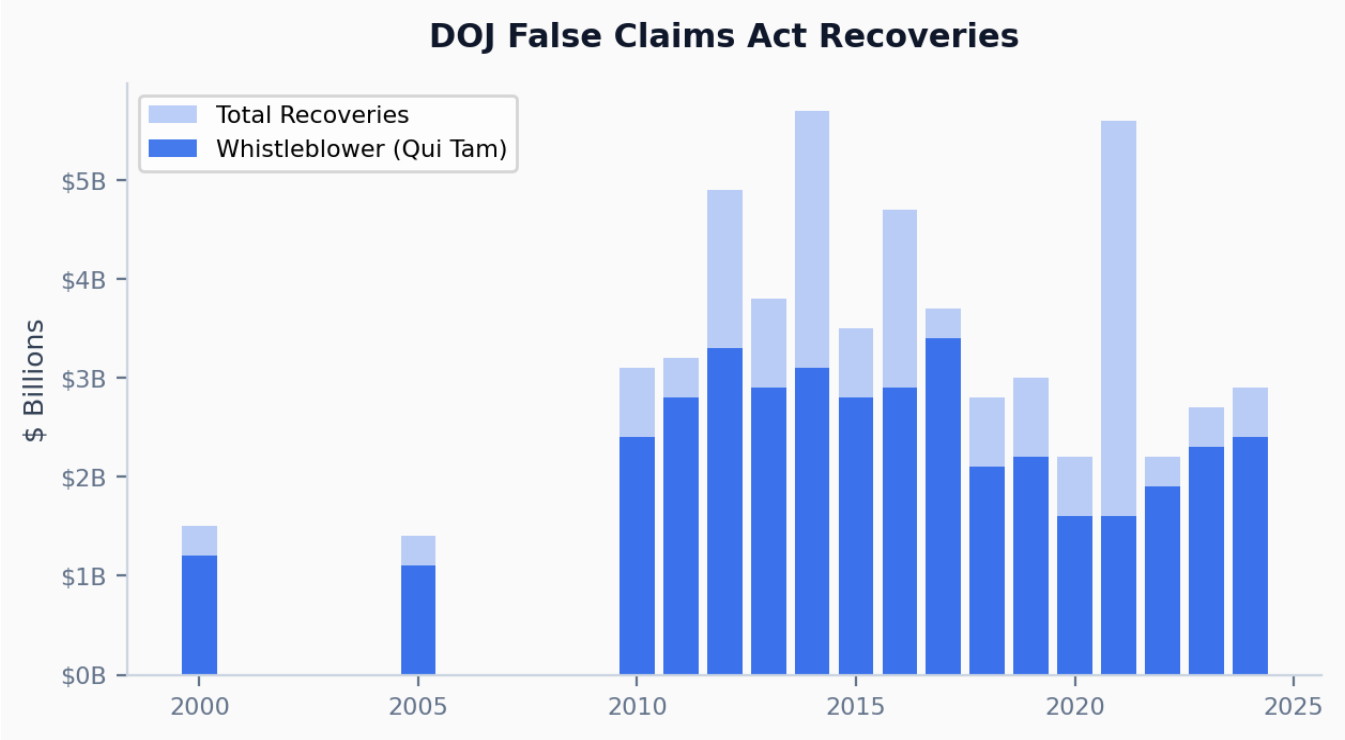
Specialty	Providers	Exclusion Rate
Legal Medicine	124	0.8%
Addiction Medicine	405	0.2%
Interventional Pain Manageme	1,615	0.2%
Pain Management	2,816	0.2%
Pathology	668	0.1%
Plastic Surgery	1,408	0.1%
Specialist	2,763	0.1%
General Practice	9,097	0.1%
Family Medicine	1,018	0.1%
Physical Medicine and Rehabi	9,043	0.1%
Preventive Medicine	1,054	0.1%
Family Practice	116,540	0.1%

Source: CMS Medicare Part D Prescribers (2023) + OIG LEIE

The Bigger Picture

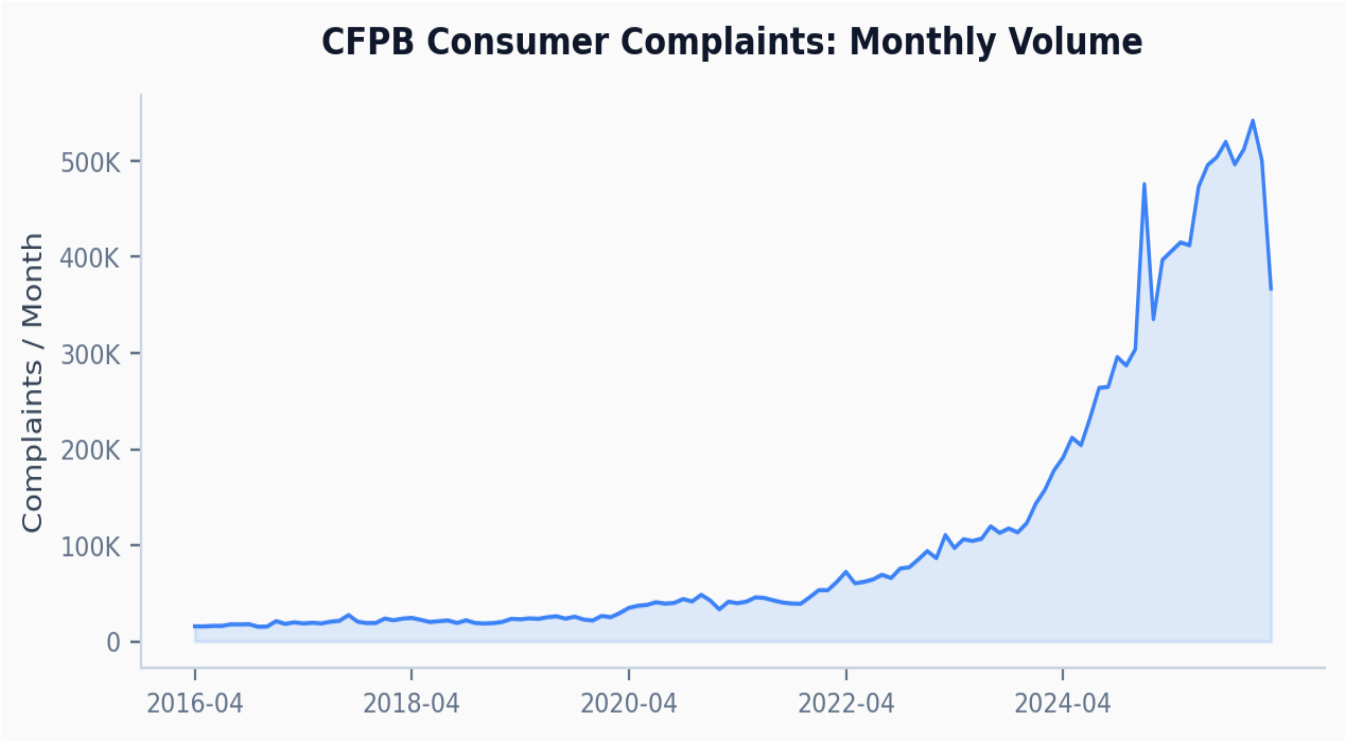
Fraud doesn't happen in isolation. Consumer complaints spike before enforcement. Whistleblower filings hit record levels. The DOJ recovered $75 billion through the False Claims Act since 1986. Here's the macro view.

False Claims Act: Record Recoveries



Source: DOJ Civil Division FCA Statistics. Qui tam = whistleblower-initiated cases.

Consumer Complaints as Early Warning



Source: CFPB Consumer Complaint Database (14M complaints, 2011-2026).

Where to Dig Deeper

This analysis surfaces patterns. The following areas are where the patterns are strongest and the public data is deepest.

PPP shared-address clusters

18,776 loans at addresses with 5+ applications. Cross-reference with business registries to find shell companies.

Round-dollar PPP loans

53% of anomalies vs 3% of normal loans are at exact $10K increments. Combined with zero employees: highest-priority fraud signal.

Beneish M-Score outliers

Cross-reference the 35 flagged companies with short-seller reports, SEC comment letters, and auditor changes.

Healthcare billing intensity

Providers 3+ standard deviations above specialty norms in claims-per-beneficiary. Compare against state medical board actions.

CFPB complaint acceleration

Companies with 3x+ monthly spikes relative to their rolling average frequently face enforcement within 12 months.

Methodology & Reproducibility

All data is from public federal sources. All analysis scripts are available at the project repository. Python 3.11+, pandas, scikit-learn, matplotlib required. Run scripts 01-05 (download) then 10-14 (analyze) to reproduce all results.

Models: PPP = Isolation Forest (unsupervised, contamination=0.02). Corporate = Beneish M-Score (formula, threshold -1.78). Healthcare = Random Forest (500 trees, balanced weights, AUC 0.67). Cross-cutting = CFPB velocity spike detection (3x rolling average).

Limitations: Statistical patterns, not confirmed fraud. PPP is unsupervised (no labels). M-Score designed for manufacturing. Healthcare limited by extreme class imbalance. CFPB reflects reporting behavior, not just actual fraud.

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